

Rafael Gonçalves

GHC 9223, Carnegie Mellon University

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Profile & Research Interests

Broadly, I am interested in **software security** and **formal methods**. I am particularly interested in static and dynamic program analysis, and in developing tools that allow developers to efficiently analyze real-world software.

Education

Carnegie Mellon University & Instituto Superior Técnico, University of Lisbon

Pittsburgh, PA & Lisbon, Portugal

PHD IN COMPUTER SCIENCE

Aug. 2024 - present

- Advised by Prof. José Fragoso Santos, Prof. Limin Jia & Prof. Pedro Adão
- Supported by a CMU Portugal Dual Degree PhD fellowship

Instituto Superior Técnico, University of Lisbon

Lisbon, Portugal

MSc IN COMPUTER SCIENCE & ENGINEERING

Sep. 2021 - Nov. 2023

- GPA: 18/20
- Specializations: Cybersecurity · Distributed Systems
- Master Thesis: Specification-Driven Synthesis of Summaries for Symbolic Execution (advised by Prof. José Fragoso Santos & Prof. Pedro Adão)

Instituto Superior Técnico, University of Lisbon

Lisbon, Portugal

BSc IN COMPUTER SCIENCE & ENGINEERING

Sep. 2018 - Jul. 2021

- GPA: 17/20

Experience

INESC-ID

Lisbon, Portugal

RESEARCH ASSISTANT

Jan. 2024 - Oct. 2024

- Group: Automated Reasoning and Software Reliability (ARSR)
- Project: "RIGA: Reasoning Over Indirect Discrimination" (DOI: 10.54499/2022.03537.PTDC)
- Supervisor: Prof. José Fragoso Santos

Carnegie Mellon University

Pittsburgh, PA

VISITING SCHOLAR

Oct. 2023 - Dec. 2023

- Worked on extending NODEMEDIC, a dynamic taint analysis tool, to detect prototype pollution vulnerabilities
- Host: Prof. Limin Jia

Publications

R. Gonçalves, F. Ramos, P. Adão, J. Fragoso Santos. Specification-Driven Generation of Summaries for Symbolic Execution. To appear in: European Symposium on Programming (ESOP 2026), Turin, Italy, 2026.

R. Gonçalves, F. Gouveia, I. Lynce, J. Fragoso Santos. Proxy Attribute Discovery in Machine Learning Datasets via Inductive Logic Programming. In: Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2025), Hamilton, Canada, 2025.

R. Gonçalves, F. Ramos, P. Adão, J. Fragoso Santos. Poster: Specification-Driven Synthesis of Summaries for Symbolic Execution. Presented at: ISSTA/ECOOP 2023, Seattle, WA, USA, 2023.

Teaching

Spring 2023, 2024 **Highly Dependable Systems (MSc)**, Teaching Assistant

Summer 2024 **Compilers (BSc)**, Teaching Assistant

Courses

Instituto Superior Técnico, University of Lisbon

BINARY ANALYSIS WITH APPLICATIONS TO MACHINE AND DEEP LEARNING

- Taught by: Prof. Michele Ianni (University of Calabria)

Lisbon, Portugal

Feb. 2023 - Mar. 2023

Honors & Awards

- 2024-29 **Dual Degree PhD Fellowship**, CMU Portugal
- 2024 **Outstanding Teaching Award**, Instituto Superior Técnico, University of Lisbon
- 2021 **Academic Excellence Diploma**, Instituto Superior Técnico, University of Lisbon
- 2018-20, 2023 **Academic Merit Diploma**, Instituto Superior Técnico, University of Lisbon

Skills

- Programming** C · Python · Haskell · JavaScript · Java · Rust
- Frameworks** Spring · gRPC · PyTorch · Flask
- Security** Binary Exploitation · Web Security · Reverse Engineering
- Languages** Portuguese (native) · English (fluent) · French (basic)